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Recharging Retail: Estimating Consumer Demand Spillovers from Electric Vehicle Charging Stations

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Abstract. *Problem definition:* We estimate the impact of electric vehicle (EV) charging stations on volumes of consumer foot traffic received by nearby retail establishments. We also explore the conditions under which any effects manifest. *Methodology/results:* We use a differences-in-differences design, exploiting the staggered introduction of Tesla Supercharger stations across the United States. We combine data on Supercharger installations with mobile phone–based estimates of retailer foot traffic. We explore heterogeneity in the treatment effect, in terms of EV charger characteristics, visitor characteristics, establishment type, and local physical context. We estimate that establishments experience an average 4% increase in monthly visits following the installation of a Tesla Supercharger. These effects arise primarily for retailers that offer relatively quick services (e.g., fast food) and for those located very near to the charger (within 150 meters). The effects are also more pronounced when the Supercharger is one of the first EV chargers introduced into the local area. *Managerial implications:* We document evidence of the positive retail demand spillovers arising from EV charging station infrastructure. We also document the conditions under which the benefits manifest. Insights for EV network operators, retailers, and policymakers are included.

Supplemental Material: The online appendix is available at <https://doi.org/10.1287/msom.2022.0519>.

Keywords: Tesla • supercharger • electric vehicles • retail partnership • omnichannel • online business

1. Introduction

Electric vehicles (EVs) have the potential to reduce greenhouse gas emissions significantly. Recent estimates indicate that EVs generate at least 25% fewer emissions than combustion-based vehicles over a vehicle’s lifespan (Woody et al. 2022). However, consumer adoption of EVs heavily depends on charging infrastructure (Zhang et al. 2016), and that infrastructure remains lacking in many areas. In California and New Jersey, for example, there are now roughly 50 EVs for every charging station (Woodyard 2022). Various stakeholders, including government, utility providers, and EV manufacturers, have sought to address this gap in recent years by investing heavily in new charging infrastructure. For example, the U.S. government has recently allocated nearly US\$5 billion toward a national electric vehicle charging network (FHA 2022), and Electrify America, a subsidiary of Volkswagen, has stated that it plans to invest \$2 billion USD in its network over the next 10 years (ElectrifyAmerica 2021).

A number of academic studies have explored the optimal allocation of subsidies and investments in charging infrastructure, with the goal of better facilitating EV adoption (Yu et al. 2022, Zhang and Dou 2022). Several

studies have examined the interplay between EV demand and charging infrastructure availability in a given geography. Importantly, however, such work frequently overlooks a potentially important stakeholder: neighboring brick-and-mortar retailers. Many EV charging stations are hosted at popular retail locations, the idea being that these locations are likely to prove convenient for consumers, enabling them to pursue other activities while they wait for a vehicle to charge. Charging station manufacturers frequently appeal to brick-and-mortar retailers to engage in hosting partnerships by making claims that such arrangements can benefit the retailer by generating foot traffic.

Tesla solicits brick-and-mortar retail hosts with claims that “Tesla charging is an opportunity to connect with a passionate demographic of drivers,” and that Superchargers attract “new and repeat customers to your location” (Tesla-Supercharger Site Hosts). Many retailers echo these claims. Kohl’s, for example, has recently announced plans to invest directly in its own charging infrastructure, claiming that customers spend approximately US\$1 for every minute of charging in a Kohl’s parking lot (Kohl’s 2021). An obvious question, however, is whether such customer spending is actually a

spillover benefit from charging infrastructure; the possibility remains that many of those consumers would visit and make purchases even in the absence of charging infrastructure. More generally, the specific conditions under which demand spillovers can be expected to manifest remain unclear. These questions are important because the answers have implications not only for retailers but also for the broader strategy of all stakeholders who might seek to expand charging infrastructure. Accordingly, we address the following research questions in this work:

*To what extent do EV charging stations drive demand spillovers to neighboring brick-and-mortar retail establishments?
Under what conditions can an establishment expect demand spillovers to emerge?*

We address these questions focusing on the Tesla Supercharger network. We combine records of Tesla Supercharger station openings with SafeGraph's estimates of consumer foot traffic visiting brick-and-mortar retail establishments between January 2018 and February 2022. Exploiting the staggered installation of Tesla Superchargers across the United States, we estimate the effect of station openings on retailer foot traffic via a difference-in-differences design. Our estimates indicate that EV charging stations yield meaningful demand spillovers for nearby retail establishments. Specifically, we estimate that a new Supercharger drives a 4% increase in monthly visitor volumes and an approximate 5% increase in monthly spending, on average. We also document important sources of heterogeneity. We show that effects manifest primarily within a relatively narrow geographic radius (under 150 meters), that they are primarily associated with "rapid" charging options, and that they are most likely to emerge when there is temporal alignment in terms of the duration of a charging session and the duration of an establishment's typical service encounter. Finally, we document important competitive dynamics, demonstrating that these benefits are subject to first-mover advantages yet that they nonetheless also attenuate as additional EV charging options enter a local area.

Our results have important implications for a variety of stakeholders. Specifically, our findings suggest that policymakers and EV manufacturers should keep brick-and-mortar retailers in mind as they consider investments and subsidies for charging station infrastructure. To the extent retailers benefit from charging infrastructure, they may have an incentive to help fund it, implying that subsidies might be better allocated elsewhere. The effect sizes and heterogeneity we document have direct implications for understanding the magnitude of interest that different retailers are likely to have in investing.

Last, our findings have important implications for the operations of brick-and-mortar retailers. Our results

demonstrate which retailers stand to benefit most from EV charging infrastructure, and they also speak to how much a particular retailer stands to lose from failing to invest early. From a theoretical standpoint, our work sheds new light on the contingent factors that can affect the manifestation of spillovers between (complementary) offline services, highlighting the importance of alignment in service encounter characteristics, namely service encounter durations and the degree of physical proximity.

2. Related Work

EVs and EV charging stations comprise a product system (Katz and Shapiro 1994) that is characterized by indirect network effects. Many electric charging stations are owned and operated by third-party service providers, so EVs may also be viewed as a multisided platform product (Boehm et al. 2020). Prior studies have argued and shown that the availability of EV charging stations is an important precursor to EV adoption (Zhang et al. 2016) and that a lack of EV charging station penetration can be a major impediment to adoption (Ma et al. 2019). This interplay reflects the well-known chicken and egg problem of platform business models (Constantinides et al. 2018).

A great deal of work has examined the network effects at play in EV markets, with the goal of informing stakeholder strategy. For example, Yu et al. (2016) considers the dynamics between EV adoption decisions and investment by third parties in charging infrastructure, concluding that a lack of coordination leads to underinvestment in charging networks, delaying the adoption of EVs. Zhang and Dou (2022) and Yu et al. (2022) explore optimal strategies for government subsidy to foster EV adoption. Zhang and Dou (2022) contrasts EV ownership subsidies with subsidies for charging station operators, concluding that the optimal strategy depends on the ratio of EV ownership and charging station supply in a particular geography. Yu et al. (2022) consider the question of how the government should approach the expansion of EV charging networks, considering per-station subsidies paid to EV manufacturers, direct station construction, or consumer vehicle purchase.

Subsidies and investments in charging infrastructure are important because the costs associated with installing and hosting an EV charging station manifest in several ways. Beyond upfront monetary investments and the physical parking space required to install a station, hosts also bear the burden of monitoring and securing physical infrastructure. Although this may sound innocuous, media reports highlight increasing rates of vandalism and theft of physical hardware, for example, charging cables cut to harvest and resell the copper as scrap metal, activities that render charging infrastructure inaccessible and lead to negative experiences for EV owners.

A common feature of prior studies in this space is that EV ownership and charging station investments are typically considered in isolation, without attention to all stakeholders' interests, for example, neighboring retailers. Such a view of the market is likely to be incomplete because brick-and-mortar retail establishments also have a vested interest in charging infrastructure. To the extent retailers value and benefit from any demand spillovers introduced by EV charging stations, prior conclusions around the optimal approach to encouraging EV adoption and charging infrastructure expansion may, in fact, be suboptimal.

Recent industry reports highlight that retailers not only host EV charging stations in anticipation of spillover benefits, but many even go so far as to subsidize the electricity (Gillera et al. 2021). For example, Hy-Vee has purposefully begun to install electric vehicle charging stations alongside all of its retail locations as part of its sustainability branding, and it has coordinated with Tesla to subsidize the charging fees for Tesla owners. McDonald's, Wawa, and other convenience, grocery, and fast-food chains have pursued similar strategies. Recent academic work also documents systematic evidence of these investments; Arlt and Astier (2023) observe that more than 50% of EV charging stations in the United States are colocated with retail businesses and that typical pricing schedules generally depart from first-best pricing schemes. Based on that observation, Arlt and Astier (2023) conclude that host retail establishments are subsidizing electricity prices in anticipation of retail demand spillovers. A natural question, of course, is whether and when those benefits actually manifest.

EV manufacturers and EV charging network operators also engage in practices that amplify the potential for retailers to benefit. Each provides EV drivers with apps and tools to help them locate EV charging infrastructure, for example, via in-car displays. Drivers are typically provided with details about a given station's power output, their vehicle's compatibility with the station hardware, and real-time stall availability. Furthermore, manufacturers often provide the driver with details on other, colocated retail establishments and services at the charging location, identifying nearby establishments a driver might visit as they wait for their vehicle to charge. An example of this is provided in Figure EC.1 in the online appendix.

3. Methods

3.1. Study Context

The prevalence of EV charging stations in the United States has risen a great deal over the last 10 years. As of 2022, there were an estimated 46,000+ public charging stations nationwide. With private initiatives like Electrify America and public EV infrastructure investments, a variety of new charger networks have been gaining

market share. That said, Tesla remains the dominant player, maintaining approximately 70% of all EV sales in 2021 (Lambert 2022).

We thus focus our analysis on Tesla. We examine the impact that a new Tesla Supercharger station has on customer foot traffic to nearby retail establishments. Although Tesla's charging network is unique in that it is a closed system, it has recently been announced that Tesla's network will soon be opened to other makes and models of EV. Furthermore, although the Supercharger network offers atypically fast charging speeds, competitor charging technology is rapidly catching up; Volkswagen's Electrify America, for example, now offers more advanced charging options in some locations. Our findings can therefore be expected to generalize to several other EV charger networks around the United States.

Tesla encourages businesses to apply for the right to host a Supercharger station, claiming that retailers will gain access to a valuable source of new and repeat customers. Tesla directly installs, manages, and operates all new Supercharger installations and requires that retailers supply several parking spaces (typically, eight spaces are required). This is not true of all EV charging networks, however; many EV charging options require that hosts bear construction and maintenance costs, although subsidies may be provided by local government. With Superchargers, hosts are typically not paid any compensation by Tesla, and each installation requires two to four weeks of on-site work. As of March 2022, 1,585 Tesla Superchargers (excluding slower "destination chargers," which are generally overnight chargers) have been installed around the United States. We collected data on the approximate opening date for each of these Superchargers, based on crowd-sourced reports on Tesla owner discussion forums.

3.2. Data

Our period of observation runs from January 2018 through February 2022. We combine data on Tesla Supercharger openings in the United States with mobile phone-based estimates of retailer foot traffic. We collected data on station openings from crowd-sourced reports in Tesla owner online forums.¹ We then scraped the Tesla Find-us website to identify the most common (20+) brick-and-mortar retail brands having establishments nearby to the Superchargers in our sample. Next, we obtained data from SafeGraph, tracking multiple measures of establishment-level consumer demand, primarily foot traffic (Holtz et al. 2020), for all U.S.-based brick-and-mortar retail establishments operated by those same brands. SafeGraph provides geospatial data derived from a large set of mobile app partners that record users' location information subject to their consent and in compliance with local privacy laws. Using geofencing, location data are aggregated by SafeGraph

to create dynamic estimates of unique visitor and visit counts, median dwell times, and spending across various physical locations of interest, including retail establishments. Brelsford et al. (2022) provide more details about SafeGraph data and report empirical validations of SafeGraph’s visit measures. Our sample of SafeGraph data for the top-mentioned brands panel ultimately includes 107,591 unique brick-and-mortar retail establishments spanning 2,892 counties and incorporates foot traffic estimates associated with 21,608,586 unique mobile devices. We plot the number of establishments in our data, by brand, in Figure 1.

In our primary analyses, we identified a specific brick-and-mortar retail location as “treated” by a Tesla Supercharger if it was located within 200 meters (this choice was based on prior literature (Smith and Butcher 2008) that determined that 200 meters was the upper bound of people’s willingness to walk from a parking structure to their destination; in robustness checks, we relax this definition of proximity). We thus have 1,340 treated retail establishments in our sample. In Table EC.1 of the online appendix, we present descriptive statistics for the complete panel of retailer-month observations associated with the 107,591 physical retail establishments. The variable $Supercharger_{i,t}$ is a binary indicator of whether a Supercharger station is actively operating in proximity to (within 200 meters of) the retail establishment; $VisitCounts_{i,t}$ is the estimate of individual visits recorded by SafeGraph at a given retail establishment, i , in a month, t . Similarly, $VisitorCounts_{i,t}$ reflects the number of unique devices recorded as having visited a given retail establishment, i , in a month, t . These are our

primary dependent variables. Using this initial sample, we next implement a matching procedure, recognizing that Supercharger construction is unlikely to be assigned at random and may very well be positioned with local amenities and retail brands in mind.

3.3. Matching

We construct our matched sample as follows. First, for each treated establishment, we identify the entire set of never-treated establishments that are operated by the same brand in the same county. Among this set of candidate controls, we identify the nearest neighbor based on (i) raw visit counts observed over each of the three months leading up to the Supercharger station’s arrival and (ii) latitude and longitude. We identify these matches employing R’s *MatchIt* package. Given the computational burden involved in this process (due to the large volume of candidate retailer pairs), we use a greedy nearest neighbor algorithm (Ho et al. 2007). This procedure results in a matched panel of 1,512 unique retailer establishments, including 756 that are eventually treated. In Figure 2, we provide two visual examples of matched pairs: one in Seattle and one in New York City. The figures depict two Supercharger stations (pins), along with their treatment radii (depicted as encompassing circles), two treated retail establishments (smaller circle markers shown within the treated radius, one Starbucks and one Dunkin’), and their matched, untreated “twins” (smaller circle markers outside the treated radius). One possible concern with this matching procedure is that we may select control locations that are actually most susceptible to demand cannibalization,

Figure 1. Volume of Establishments by Brand Among Top-Mentioned Brands

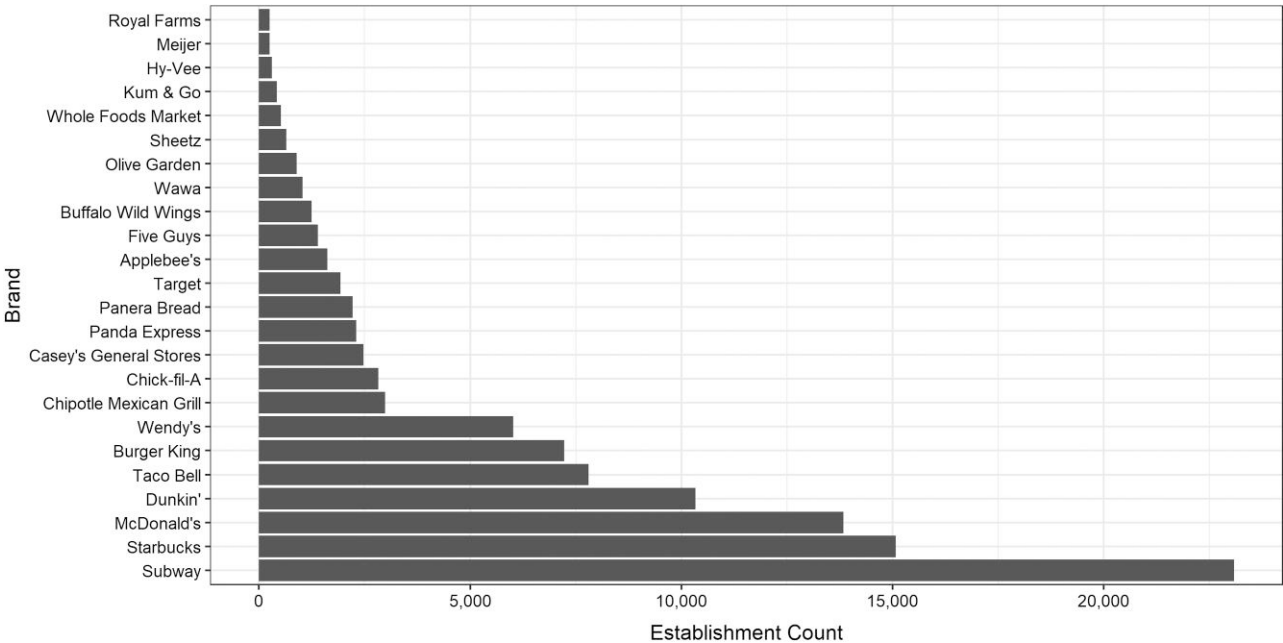


Figure 2. (Color online) Sample Treated (within the treated radius) and Control (outside the treated radius) Matched Locations by Brand



Notes. (a) Matched Starbucks in Seattle. (b) Matched Dunkin's in NYC.

leading to a violation of the stable unit treatment value assumption (SUTVA). We address this possibility in later analyses, reported in Section 4.4, where we change our unit of analysis to the brand-county. In that section, we also explore other aspects of competition around EV charging station installations. It is worth noting that our use of a matched sample throughout the study implies that our estimates reflect local average treatment effects (LATEs).

In Table 1, we report descriptive statistics for our matched panel. Correlations among our variables are presented in Table EC.2 in the online appendix. Besides the visit and visitor counts, we also obtain a number of other measures describing other aspects of the visits and visitors to each retail establishment, including $AvgDistFromHome_{i,t}$, which indicates the average distance in a given month between the establishment's location and the census-block groups where its visitors reside, and $MedianDwell_{i,t}$, which reflects the median

duration of visits to the establishment in a given month. Finally, we obtain measures of monthly labor force size, and the unemployment rate in the county where each establishment is located from the Bureau of Labor Statistics.

3.4. Empirical Specification

Our identification strategy exploits the staggered treatment of different physical retail establishments over time, that is, the nearby construction of new Tesla Supercharger stations. We thus estimate a staggered difference-in-differences (DID) regression of the following form, considering a variety of demand measures, including $VisitorCounts_{i,t}$, $VisitCounts_{i,t}$, $MedianDwell_{i,t}$ and $\log(AvgDistHome_{i,t})$. Later, we will also consider data on actual customer spend, available for a subsample of establishments, that is, $\log(TotalMonthlySpend_{i,t})$ and $MedianSpendPerTransaction_{i,t}$. The binary indicator *Supercharger* shifts from zero to one for treated establishments

Table 1. Summary Statistics of the Monthly Matched Panel

Statistic	N	Mean	Standard deviation	Minimum	Maximum
Visits	74,358	797.37	944.10	1	41,527
Visitors	74,358	535.17	596.78	1	24,449
DistFromHome (km) ^a	73,568	21.31	62.86	0.04	4,087.54
MedianDwellTime (min)	74,358	19.92	52.19	4.00	1,425.00
Supercharger	74,358	0.20	0.40	0	1
LaborForce	74,358	590,101.50	864,921.80	5,858	5,265,515
Unemployment (%)	74,358	5.21	3.09	0.70	32.70

^aFor privacy reasons, SafeGraph does not report home CBG-based metrics in cases where fewer than four visitors have arrived from that CBG in that period. We therefore have some missing values in the DistFrom Home measure.

when a Tesla Supercharger opens for service nearby:

$$Y_{i,t} = \beta \cdot \text{Supercharger}_{i,t} + \gamma \cdot X_{i,t} + \mu_i + \eta_{s,t} + \varepsilon_{i,t}. \quad (1)$$

All regressions are performed on the matched panel described earlier. In addition to retail establishment fixed effects μ_i , we include absolute (rather than seasonal) state-month-year fixed effects, which account for state-specific unobserved temporal trends, for example, state policy responses in the handling of COVID-19.

We subsequently report a series of robustness checks in Section 4.3, addressing concerns of self-selection on treatment, sample selection, and sensitivity to the choice of estimator. We then explore competitive dynamics in Section 4.4, and we verify the mechanisms underlying the observed effects. For example, we explore how the effects are moderated by competing EV charging options in the local area, and we provide evidence that effects are contingent on the volume of Tesla owners who live in the region (although we acknowledge that our measure of Tesla ownership volumes is very likely to be endogenous, given that EV ownership volumes and EV charging options influence one another).

Last, we explore heterogeneity in the treatment effects based on the characteristics of the retail location, its context, and its historical visit activity. In particular, we consider the number of other complementary amenities nearby, the importance of parking availability, and the urban density of a location. Furthermore, we explore more flexible definitions of treatment. We consider how treatment effects vary over different ranges of distance between the charging station and a retailer, and we explore how treatment effects vary depending on the alignment between a retailer's typical service encounter duration and the duration of an EV charging session.

4. Results

4.1. Effect on Visitor Volumes

In Table 2, Panel A, we estimate the impact of a Supercharger introduction on the number of distinct visitors arriving at a proximate retail establishment every month. We report our primary results using ordinary least squares (OLS) regression in column 1. In column 2, we repeat our estimation using Poisson regression, cognizant

of the nonnegative count nature of the outcome variable. Each estimation yields a positive, statistically significant effect on visitor volumes per month. The Poisson regression has the intuitive benefit of yielding coefficients that are interpretable as elasticities. The coefficient associated with Supercharger presence suggests that, on average, Superchargers yield an approximate 4% increase in monthly visitors ($e^{0.04} - 1$).

Next, we divide treatments into two categories, distinguishing between rapid chargers (Panel B) and slower (72 kWh) urban chargers (Panel C). We make this distinction for two reasons. First, substantial differences exist in the time required to complete a full charge, which may have implications for the synergies between these Superchargers and typical retail trips that visitors might wish to undertake as they wait. Urban Chargers can take between 45 minutes and one hour to complete a full vehicle charge (Lambert 2017) and tend to be located in denser neighborhoods. By contrast, rapid chargers (which supply electricity at rates of 120–250 kWh) yield a typical charge time of 20–30 minutes and are typically located along motorways or in suburban areas. Rapid chargers are typically focused on supporting commuter traffic.

Second, urban chargers tend to be surrounded by many more retail establishments. As a result, the demand benefits they afford are likely to diffuse across a wider set of establishments; that is, EV owners will not be a particularly captive audience for any one retailer. Third and last, urban chargers are often hosted within gated parking structures, many of which require additional paid access. This payment implies that such chargers are generally more expensive. Considering the two types of chargers separately, we find results consistent with our expectations. Specifically, compared with the general estimate, we find a larger, more positive effect estimate from rapid chargers (columns 1 and 2, Panel B). In contrast, when we focus on urban chargers, we continue to see a positive coefficient, but it ceases to be statistically significant (columns 1 and 2, Panel C).

4.1.1. Impact on Visits. It is important to assess whether the increased visitor counts also translate to

Table 2. Impact of Superchargers on Monthly Visits and Visitors per Establishment (Within 200 Meters of a Supercharger)

	Visitors		Visits	Visits/Visitors
	(1) OLS	(2) Poisson	(3) OLS	(4) OLS
Panel A: All chargers				
Supercharger	24.57* (10.48)	0.04* (0.02)	35.93* (17.13)	−0.01* (0.01)
Log(LaborForce)	495.88** (166.33)	0.93** (0.33)	660.80** (247.35)	−0.13 (0.09)
Unemployment	−24.14*** (3.38)	−0.05*** (0.01)	−34.48*** (4.80)	0.00 (0.00)
Fixed effects	Establishment, State-Month-Year			
No. of observations	74,358	74,358	74,358	74,358
Establishments	1,512	1,512	1,512	1,512
Adjusted R^2 (full model)	0.86		0.85	0.49
Adjusted R^2 (proj model)	0.01		0.01	0.00
Pseudo- R^2		0.90		
Panel B: Rapid chargers				
Supercharger	30.02** (10.62)	0.04* (0.02)	41.27* (18.19)	−0.02* (0.01)
Log(LaborForce)	387.76* (153.71)	0.52* (0.27)	486.67* (233.51)	−0.18+ (0.10)
Unemployment	−16.49*** (2.45)	−0.04*** (0.00)	−23.06*** (3.59)	0.00 (0.00)
Fixed effects	Establishment, State-Month-Year			
No. of observations	61,017	61,017	61,017	61,017
Establishments	1,240	1,240	1,240	1,240
Adjusted R^2 (full model)	0.86		0.86	0.51
Adjusted R^2 (proj model)	0.01		0.00	0.00
Pseudo- R^2		0.90		
Panel C: Urban chargers				
Supercharger	8.59 (33.59)	0.07 (0.08)	20.66 (49.80)	0.00 (0.02)
Log(LaborForce)	−824.74* (412.44)	2.05* (0.80)	−969.78 (661.63)	0.24 (0.58)
Unemployment	−32.00** (12.08)	−0.02 (0.02)	−45.50** (16.44)	0.00 (0.01)
Fixed effects	Establishment, State-Month-Year			
No. of observations	13,341	13,341	13,341	13,341
Establishments	272	272	272	272
Adjusted R^2 (full model)	0.84		0.83	0.45
Adjusted R^2 (proj model)	0.01		0.01	−0.00
Pseudo- R^2		0.91		

Note. SEs clustered by establishment in parentheses.
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

greater overall visit volumes to ensure these new visitors are not displacing established customers at the intensive margin. We therefore examine the impact of Supercharger installations on total monthly visit counts, as reported in column 3 of Table 2, across all subtables. Our results demonstrate that a Supercharger significantly increases total monthly visits to nearby establishments, particularly in the case of rapid chargers. Superchargers thus drive a net increase in both visitors and visits.

4.1.2. Visits per Visitor. Last, in column 4 of all panels, we examine the impact of a Supercharger’s introduction on the number of monthly visits per unique visitor. We observe a significant reduction from chargers in general ($\beta = -0.01, p < 0.05$), and from rapid chargers in particular ($\beta = -0.02, p < 0.05$). This result indicates that the new customers who are drawn to visit EV charging stations tend to visit less frequently than the established customer base. This finding is expected, given the primary

Table 3. Impacts on Monthly Spending and Monthly Transactions per Establishment for Matched Panel (200 Meters of SCs), Top-Mentioned Brands

	Log(TotalMonthlySpend)	MedianSpendPerTransaction
Supercharger	0.05*** (0.01)	0.56*** (0.11)
Log(LaborForce)	1.35*** (0.29)	−0.95 (2.59)
Unemployment(%)	−0.04*** (0.01)	−0.05 (0.04)
No. of observations	25,714	25,714
No. of establishments	1,403	1,403
Adjusted R^2 (full model)	0.93	0.91
Adjusted R^2 (proj model)	0.01	0.00

Note. SEs clustered by establishment in parentheses.
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

reason for their visit is likely a desire to charge their vehicle, whereas shopping is likely secondary.

4.2. Effect on Visitor Characteristics

4.2.1. Spend. It is also important to assess the impact of EV chargers on actual revenue earned by retail establishments. Business owners may be concerned that new EV owner visits will crowd out historical customers without material sales benefit. Accordingly, we assess the effects on transaction spending, using SafeGraph’s spend data set. As with visitation data, transaction spending is reported monthly for each establishment. However, spend data are only available for a subset of establishments, namely those that partner with payment processors that sell their data to SafeGraph. Using this sample, we consider two additional outcome measures: Log(*TotalMonthlySpend*) and *MedianSpendPerTransaction*. The descriptive statistics for these variables are reported in the online appendix, in Table EC.3, and the results of our estimations are reported in Table 3. We note about a 5% increase in monthly spending at treated locations and an increase in spending per transaction, signaling an increase in wealthier clientele.

4.2.2. Visit Duration. It is possible that the customers that a Tesla Supercharger attracts spend less or more time than a typical consumer would, for example, because the duration of their charging session binds them. This is another important consideration, as it would directly determine how much business the establishment can expect from EV-charging customers. We thus consider SafeGraph’s measure of monthly “Median Dwell Time,” an estimate of the median time a visitor lingers at the retailer. Using this dependent variable (DV), we obtain the estimates reported in column 1 of Table 4, where no significant changes are noted.

4.2.3. Distance Traveled. The introduction of a Tesla Supercharger may also affect the mix of consumers that arrive at nearby retail establishments. In particular, we might expect to see a shift in the distance that visitors have traveled. To test this, we replace the dependent variable in Equation (1) with the log of the average distance between visitors’ home census block group (CBG) and the establishment’s census block group. The results of this estimation are reported in column 2 of Table 4. We observe no statistically significant effects.

Table 4. Impact of Supercharger (All) Introduction on Other Outcomes

	Median Dwell (min) (1)	Log(DistanceFromHome) (km) (2)	AvgInc (1000 s) (3)
Supercharger	0.066 (1.091)	0.008 (0.011)	0.390* (0.160)
Log(LaborForce)	−15.162 (22.548)	0.172 (0.217)	1.136 (2.462)
Unemployment(%)	0.229 (0.392)	−0.010* (0.004)	−0.071 (0.049)
Fixed effects	Establishment, State-Month-Year		
No. of observations	74,552	73,749	73,840
Adjusted R^2 (full model)	0.454	0.884	0.898
Adjusted R^2 (proj model)	0.000	0.001	0.000

Notes. SEs clustered by establishment in parentheses. Home Census Block Group information is not always available in SafeGraph data, for privacy reasons, leading to fewer observations in column (3).
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

4.2.4. Average Income. The Tesla customer profile is rather unique in several respects. Most notably, Tesla owners (and EV owners more generally) are disproportionately high-income individuals. As of March 2022, an entry-level Tesla Model 3 has a list price of \$44,900, whereas the flagship Model S has a list price of \$94,990 (gobankingrates 2022: How much is a Tesla?). The average owner of a Tesla Model 3 has a household income more than double that of the median U.S. household (Automotive Market Research 2022).

Although we lack explicit data on each visitor's income, we can examine the income distribution in a visitor's home CBG. Using census data, we measure the average income of visitors arriving at a retailer in a given month. We first calculate the number of individuals arriving from each CBG. Then, using the relative volume of visitors across census block groups, we calculate a weighted average of CBG-median incomes (in 1,000s of dollars, as listed in Table B19013 of the U.S. Census). Taking this value as our dependent variable, we repeat our estimation and obtain the results presented in column 3 of Table 4. As expected, we find that average household income is significantly higher among visitors to a treated retail establishment ($\beta = 0.39, p < 0.05$). Although this estimate translates to an increase of just \$385 in median annual household income, this is not altogether surprising given our earlier estimate implies that the newly arriving Tesla owners will comprise a small, single-digit percentage of overall visitor traffic.

4.3. Robustness

Our matching approach ensures that, for each treated establishment, we first identify a valid control. However, the possibility remains that some other source of selection into treatment may exist. Presumably, if a selection mechanism is at play that also associates with shifts in visitor traffic, we would not necessarily expect its influence to coincide perfectly with a Supercharger's arrival. That is, if a process of self-selection drives our effects, we might then expect to observe delays in the onset of the treatment effects, or we might expect to observe onset prior to treatment. However, we must also keep in mind that the crowd-sourced nature of our data on Supercharger introductions implies that our recorded timing of treatment may be systematically delayed, as consumers reporting new stations may not have observed and reported their opening immediately. Nonetheless, we take some additional steps to help ensure our main findings are not driven by self-selection.

4.3.1. Robustness to Site Selection Based on Nearest Establishment. If retailers dictate the construction of Superchargers, determining their placement, it is possible that Superchargers arrive systematically in locations that are more likely to benefit the colocated retail establishments. Because, in our study period, customers fully

pay for charging rather than any retailer, their direct influence on installation decisions is unlikely. However, assuming retailers in our sample are actively influencing Supercharger arrivals, the most likely establishment to have engaged in this behavior is presumably that establishment located closest to the Supercharger. Accordingly, we can rule out this source of self-selection by omitting those establishments from our estimation sample. We therefore repeat our estimations, dropping those retail establishments that are located closest to the treating Supercharger (along with their matched controls). We report results with this modified sample in the online appendix, in Table EC.4, where we find that our main results remain unchanged. This finding suggests that a selected sample of establishments does not drive our findings.

4.3.2. Exploring Larger Set of Brands and Wider Treatment Radius. Our choice of brands in the primary analyses was based on those identified as the most commonly mentioned brands around U.S. Superchargers, as reported on the Tesla website. A second concern, again related to self-selection, is that some of these retail brands may have formed a partnership with Tesla.² A second related concern is that our focus on a relatively narrow radius may influence our estimates of treatment effects; establishments located further away may have experienced benefits also.

To address both concerns, we repeat the main analysis, reflected by Equation (1), but using a larger, expanded sample of establishments, implementing the same matching procedure. We supplement our original panel with additional establishments, including all branded retail establishments (North American Industry Classification System (NAICS) codes: 44-45 [Retail Trade], 72 [Food Services], and 81 [Other Services]), including treated establishments located up to 750 meters from a Supercharger. The results of our estimations with this larger panel are presented in Table 5. Maintaining the original (narrow) radius of 200 meters, we continue to observe statistically significant average effects on both visitors and visits. Considering a wider treatment radius of 750 meters, we continue to see statistically significant effects, although the effect sizes become smaller, as we would expect. These estimates demonstrate that our results are not limited to brands that were identified by Tesla directly in its app or on its website, nor are they driven by our decision to focus on a relatively narrow radius of treatment. Later, in Section 4.5, we explore the effects of physical distance more flexibly.

4.3.3. Inverse Propensity for Treatment Weighting. Although our matched sample seeks to identify the ideal control for each treated establishment, it does so in a relatively coarse manner and uses a greedy matching algorithm. It is therefore possible that systematic differences may remain between, for example,

Table 5. Impact of SC Opening on Establishments (All Brands)

	Within 750 meters				Within 200 meters			
	All SCs		Rapid SCs		All SCs		Rapid SCs	
	Visitors (1)	Visits (2)	Visitors (3)	Visits (4)	Visitors (5)	Visits (6)	Visitors (7)	Visits (8)
Supercharger	14.94** (5.09)	24.40** (8.32)	22.21*** (5.31)	34.93*** (8.95)	21.12* (8.84)	31.66* (14.38)	30.45*** (9.05)	43.84** (15.28)
Log(LaborForce)	479.78*** (94.11)	673.99*** (140.93)	353.14*** (88.83)	449.54** (141.04)	370.54** (139.31)	455.11* (208.48)	243.42+ (131.67)	246.42 (201.94)
Unemployment(%)	−17.42*** (1.31)	−26.73*** (2.11)	−16.57*** (1.33)	−25.52*** (2.29)	−21.69*** (2.39)	−31.08*** (3.49)	−17.99*** (2.11)	−25.67*** (3.14)
Fixed effects	Establishment, State-Month-Year							
No. of observations	236,793	236,793	199,638	199,638	99,735	99,735	83,071	83,071
No. of establishments	4,973	4,973	4,202	4,202	2,056	2,056	1,714	1,714
Adjusted R ² (full model)	0.84	0.83	0.85	0.84	0.86	0.86	0.86	0.87
Adjusted R ² (proj model)	0.01	0.01	0.01	0.00	0.01	0.01	0.01	0.01

Note. SEs clustered by establishment in parentheses.
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

a McDonald’s that is treated and a second that is not. We use inverse propensity for treatment weighting (IPTW) to reweight the matched sample, encouraging greater balance on observable covariates. We use covariate-balancing propensity scores (CBPSs) (Imai and Ratkovic 2014) for this purpose, using our focal estimation sample and rapid superchargers. Our propensity score function models treatment as a function of latitude, longitude, colocation with a parking lot (a binary indicator), the count of local amenities nearby reported by Open Street Maps (OSM), distance from the city center, and three lags of the primary DV. Reweighting observations in the matched sample based on the inverse of the resulting propensity scores. We repeat our estimation using the reweighted sample and obtain the results in Table EC.6 in the online appendix. Once again, all our main findings persist.

4.3.4. Treatment Effect Dynamics. To assess the dynamics of treatment effects, we use the matrix completion estimator of Athey et al. (2021), implemented within R’s *FECT* package, using an event study specification. This estimator also has the benefit of being robust to recent concerns that have been raised in the econometrics literature (Callaway and Sant’Anna 2021, Goodman-Bacon 2021) about the potential bias of the two-way fixed effect (TWFE) estimator under staggered treatments. Accordingly, we also leverage it to repeat our estimation of the average effect.

The matrix completion estimator imputes counterfactual outcomes for treated panels based on matrix factorization, that is, singular value decomposition (SVD). The matrix completion estimate of the average effect of treatment in the rapid charger panel is 31.16 ($p = 0.05$). Notably, this estimate is very similar to that we obtained using two-way fixed effect estimation in Table 2, column 1, suggesting that the TWFE estimator is relatively unbiased.

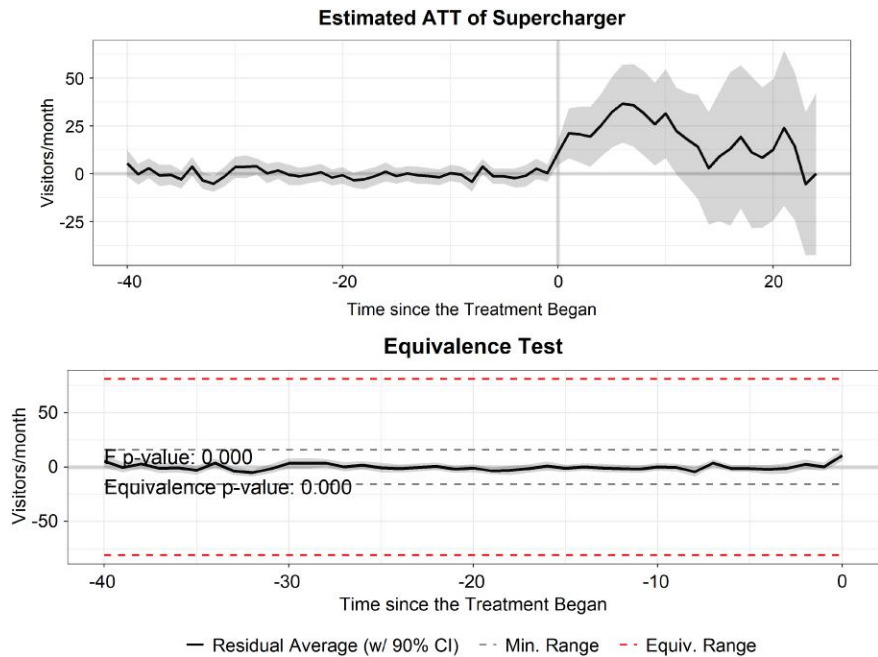
We plot dynamic estimates in Figure 3 (top) and the corresponding equivalence test for the pretreatment period in Figure 3 (bottom), which formally evaluates the parallel trends assumption. We found no evidence of statistically significant effects over the pretreatment period, consistent with the assumption of parallel trends. Considering posttreatment dynamics, we observed that the effect peaks approximately 9–10 months after treatment and then begins to subside, although with wide variance.

That the persistence of the effects is somewhat unclear beyond one year of treatment³ may be explained at least in part by competition effects, for example, alternative charging stations entering the area over time. We explore that possibility in Section 4.4.

4.3.5. Staggered DiD Robustness. In addition to matrix completion, we also consider several estimators that have recently been developed to address issues of potential bias with TWFE estimation in the presence of staggered treatments (Goodman-Bacon 2021). These estimators address the specific issue of “forbidden comparisons,” wherein previously treated units may serve as controls for later treated units, despite that the treatment effects experienced by the former may continue to evolve dynamically and may have yet to stabilize (Roth et al. 2022).

We present the event study estimates from each of six different estimators in Figure 4 based on our focal panel. These estimates are obtained using the *did2s* package in R of Butts and Gardner (2021). In every case, we find that the main effect is significant and positive, typically peaking by the 10th month after Supercharger construction. Once again, in every case, we observe pretreatment estimates consistent with the assumption of parallel trends. Depending on our choice of estimator, we also find that the treatment effects persist (Borusyak et al. 2021, Gardner

Figure 3. (Color online) Matrix Completion Estimations: ATT (Top) and Equivalence Test (Bottom)

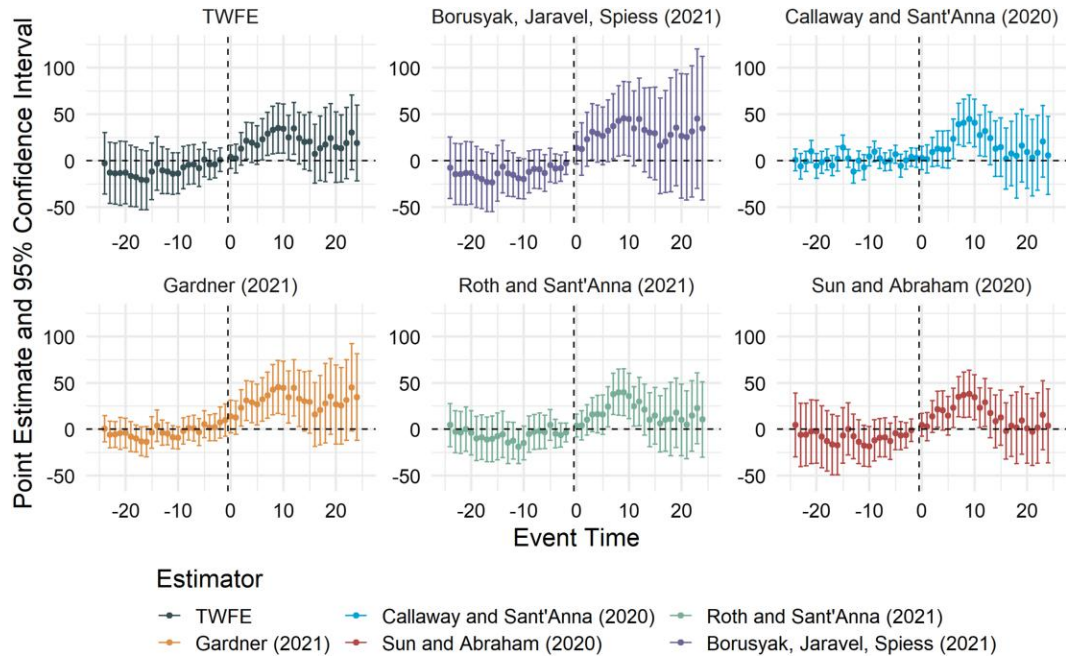


2022), although with increasingly wide standard errors, which results in a weakening of statistical significance.

4.4. Competitive Effects

As noted earlier, the long-run effects on establishment visit volumes may decline with the entry of additional charging stations in the broader region. We explicitly test this idea to understand how demand benefits from EV charging stations may depend on (i) the volume of alternative EV charging options in a region and (ii) the sequence of EV charging station introductions in regard to possible first-mover advantages, for example, perhaps

Figure 4. (Color online) Temporal Dynamics of the Treatment Effect of Superchargers on Monthly Visitor Counts



Notes. Because the time of Tesla Supercharger arrival is reported online by users when they first notice a charger becoming operational, the actual dates of charger introduction may be two to three months earlier than reported. To account for this, we perform all relative-time estimations using the third month prior to Supercharger entry as the reference period for comparison.

EV owners form habits in terms of their choice of charging locations, which hold after new stations emerge. We estimate a set of moderated regressions to understand these effects, reported in the online appendix in Table EC.8. We observe, as expected, that Superchargers' benefits are attenuated in the presence of more charging alternatives in the same county. Furthermore, we observe first-mover advantages, such that the benefits of a Supercharger are systematically larger for those that are built earlier (e.g., the first construction in the county).

The previous points also raise the possibility that the positive effects we estimate reflect a combination of both increased demand at Supercharger-proximate retailers *and* reduced demand at nonproximate retailers due to cannibalization. Was this dynamic at play, it would affect our interpretation of the treatment effect; that is, it would capture the net competitive benefit rather than a strict demand increase at the treated retailer. To explicitly test whether and to what extent demand cannibalization contributes to our estimates, we aggregate visit activity across retail establishments to the brand-county level. We modified Equation (1), modeling total monthly visitors across retail establishments for a given brand in a given county as a function of the number of treated brand establishments in the county, for example, the count of the brand's establishments in the county that were located within 200 meters of a Supercharger (*EstabsNearRapidSCs*). The results of this estimation are presented in Table EC.7 of the online appendix. We found that as more Superchargers are constructed alongside a brand's establishments in a county, this translates to net increases in county-level customer visits. This result indicates that Superchargers are not driving a simple redistribution of visitors between different establishments. We also repeated this exercise at the NAICS code county level (to allow for the possibility of cross-brand substitution), and the results remained similar. We saw a net increase in total customer visits to county establishments of a particular type as more Superchargers appeared. Ultimately, this result does not completely rule out the role of demand redistribution and cannibalization, but it does indicate that a large component of the effects we have found is driven by net-new demand, that is, an expansion of the pie.

4.4.1. Mechanism. We next seek to validate mechanisms. We assess whether the results we observe are, in fact, attributable to increased visits from Tesla vehicle owners, as we would expect. We perform three additional tests to evaluate this.

First, we consider the moderating influence of the volume of registered Teslas in a retail establishment's county over time, recognizing that analyses employing this measure will be imperfect for at least two reasons. First, the measure is a noisy proxy for market potential given that many Tesla owners may, of course, arrive from another county, particularly commuters or drivers

on long-distance road trips. Second, as noted in the Introduction, ample work makes the point that EV ownership and EV charging station volumes are code-terminated and thus subject to reverse causality. Nonetheless, we gathered EV registration data for 120 counties across 13 states from Atlas EV Hub. We used these data to calculate county-month counts of active Tesla registrations. Because the data set provides snapshots of EV registration rather than a real-time measure, we used the *akima* package in R to interpolate estimates of monthly Tesla volumes per county per month. We observed 827 registered Teslas in an average county month.

We incorporate this measure as a moderator and repeat our main regression analysis, using our primary panel of establishments among the top-mentioned brands and focusing on rapid Superchargers. We include quartile dummies for the measure, taking the bottom quartile, or fewer than four Teslas, as the reference condition. The results of this estimation are reported in Table EC.9 in the online appendix. As we expect, the effect of Supercharger installation on nearby retail establishment visit volumes manifests conditional on Tesla ownership volumes and only when a sufficiently large number of Teslas is present on the road. That said, we observe a statistically insignificant effect at the highest quartile, perhaps because higher Tesla ownership volumes correlate positively with the urban density of a location (and thus a less captive audience for the treated retail establishment). Alternatively, as noted earlier, insignificance at the upper quartile may be due in some way to the likely endogeneity of the Tesla ownership measure.

Second, we consider the features of the visitors themselves. SafeGraph distinguishes visitors based on their mobile device operating system: Android versus iOS. Small sample surveys and anecdotal evidence indicate that Tesla users are also often Apple product users, generally being consumers of higher-ticket luxury items. Indeed, the chief executive officer (CEO) of Porsche has previously gone on record noting that 91% of Porsche owners are iPhone users. iPhone users are also typically in a significantly higher income bracket, lending credence to the idea that visits by more Tesla owners should translate to more visits by iPhone owners. We should thus expect to see differential effects of Supercharger installation on iOS versus Android user visits.

SafeGraph also distinguishes visits by day of the week. This is notable because Tesla Superchargers are often seen as most beneficial for drivers on longer-distance, weekend road trips (versus shorter, daily commutes) because in the latter case, at-home or regular EV charging options, despite being slow, are often sufficient to cover charging needs. Furthermore, individuals are likely to prefer using a Supercharger when they have the freedom to make a 30- to 45-minute stop. Individuals are less likely to make use of a Supercharger during the work day, as this may necessitate leaving work to charge

Table 6. Summary Statistics of Moderating Characteristics in the Panel

Statistic	N	Mean	Standard deviation	Minimum	Maximum
PastMedianDwellTime	30,567 ^a	16.171	22.574	5.000	747.571
DistFromCharger (m)	30,567 ^a	106.764	53.367	3.847	199.961
Amenities	30,567 ^a	1.568	1.658	0	8
DistFromCityCntr (km)	61,107	13.623	10.860	0.261	83.498
InOSMParking	61,107	0.20	0.40	0	1
ParkingLotArea (sq. feet)	12,086 ^b	19,791.25	16,870.46	118.20	92,349.53

^aDefined only for treated establishments.
^bDefined only for establishments proximate to a charger within a parking lot.

and then returning. Consistent with these ideas, an anecdotal inspection of Google Places profile data for Superchargers indicates that visits are more heavily concentrated on weekends. Based on this, we would expect to see a stronger effect on retail establishment visits on weekends.

We thus test for heterogeneity in users’ phone operating system and the day of the week associated with visits. To achieve this, we modify our panel to create duplicate observations per establishment-month, splitting our dependent variables based on visitor operating system (Android versus iOS) and again based on weekend versus weekday visits. Using this modified panel, we re-estimate Equation (1), incorporating a dummy reflecting whether the observation pertains to weekends (versus weekdays) or iOS (versus Android) visitors. We interact the dummy with the treatment indicator, such that any significant moderating effect will indicate that treatment effects differ meaningfully across visit and

visitor types. The results of these analyses are presented in the online appendix, in Table EC.10. As expected, we find that effects are systematically larger among iOS users ($\beta = 26.14, p < 0.01$) and on weekend visits ($\beta = 7.37, p < 0.001$).

4.5. Heterogeneity Analyses

We now explore other sources of heterogeneity in the treatment effects. Specifically, we implement additional moderated regression analyses, once again focusing our attention on rapid Superchargers given our prior observation that those chargers drive the majority of effects. Summary statistics for the various moderators we consider are reported in Table 6.

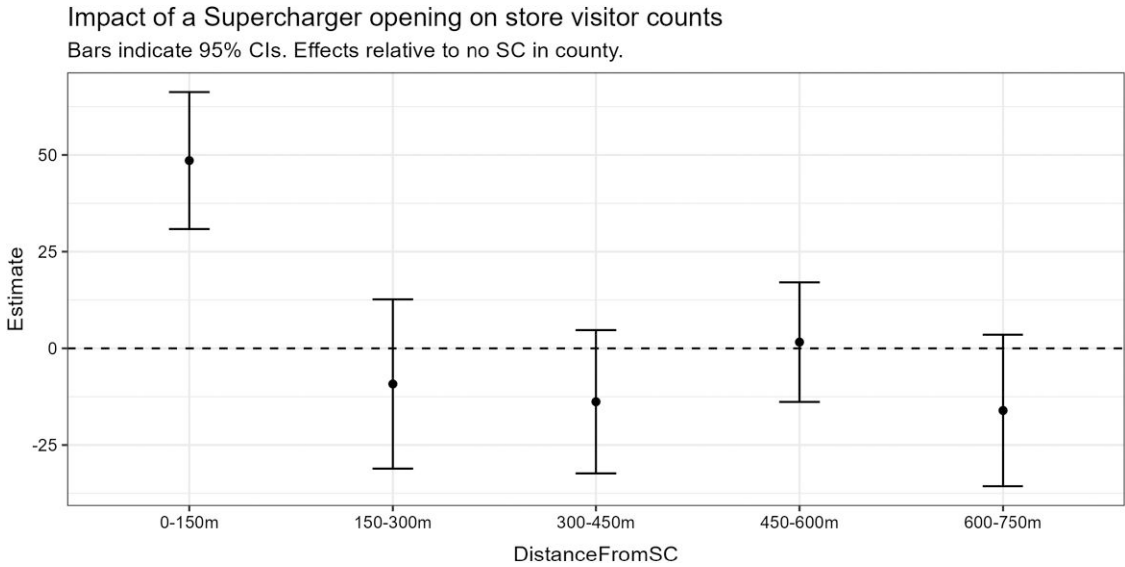
In Table 7, we present our moderated regressions. Columns 1–7 incorporate individual moderators, whereas column 7 incorporates all moderators jointly. We focus on column 7 when drawing interpretations. Our first moderator of interest is *DistFromCharger_{it}*, the distance

Table 7. Heterogeneity in the Impact of Rapid Superchargers on Establishment Visits

Dependent variable = <i>Visitors</i>	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
Supercharger	30.90** (10.68)	139.86*** (26.38)	20.45 (12.45)	22.72 (14.74)	52.83*** (14.18)	52.78*** (14.18)	129.30*** (28.89)
Log(LaborForce)	386.31* (153.69)	416.28** (152.09)	379.58* (153.73)	383.29* (153.67)	365.21* (153.69)	365.56* (153.66)	391.78* (152.38)
Unemployment(%)	−16.88*** (2.55)	−16.84*** (2.57)	−16.82*** (2.55)	−16.93*** (2.55)	−17.03*** (2.55)	−17.04*** (2.55)	−16.92*** (2.57)
Supercharger × DistFromCharger(Mts)		−1.02*** (0.20)					−1.00*** (0.20)
Supercharger × Amenities			7.70 (4.93)				9.60* (4.75)
Supercharger × DistFromCityCntr(Kms)				0.57 (0.67)			0.58 (0.64)
Supercharger × InOSMParking					−55.29** (19.47)	−56.88** (19.46)	−37.32* (18.20)
Supercharger × InOSMParking × ParkingLotArea (1000 s) sq.ft.						0.58 (0.57)	1.18* (0.57)
Fixed effects	Establishment, State-Month-Year						
No. of observations	61,107	61,107	61,107	61,107	61,107	61,107	61,107
Adjusted R ² (full model)	0.86	0.86	0.86	0.86	0.86	0.86	0.86
Adjusted R ² (proj model)	0.01	0.01	0.01	0.01	0.01	0.01	0.01

Note. SEs clustered by establishment in parentheses.
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; + $p < 0.1$.

Figure 5. Impact of Supercharger on Visitor Counts for All Brands, Matched Panel Within 750 Meters of a Charger at Distances Relative to the Charger



between the brick-and-mortar retail location, i , and its Supercharger (in meters). Note that $DistFromCharger_i$ is undefined for untreated retail locations; that said, any arbitrary value can be imputed for untreated locations, as the establishment-fixed effect will subsume it in any event. As expected, we find that retailers experience more benefits when they are located more closely to a Supercharger ($\beta = -1, p < 0.001$). We explore the relationship between Supercharger distance and treatment effect magnitude more flexibly later, considering effects at different distance ranges. Those effects are summarized visually in Figure 5.

When choosing among different Superchargers in a given city, Tesla owners might also be influenced by the availability of other complementary amenities that can be accessed while waiting for a charging session to complete beyond a focal establishment. To test this idea, we construct a measure reflecting the count of nearby amenities located within 500 meters of the retailer based on OpenStreetMap’s API. The amenities we consider include *police station, pharmacy, bank, clinic, cinema, car wash, water fountain, post office, and gas station*. Using this measure as a moderator, we find that the effect of a Supercharger on retail establishment visits is larger in the presence of other nearby amenities ($\beta = 9.6, p < 0.05$).

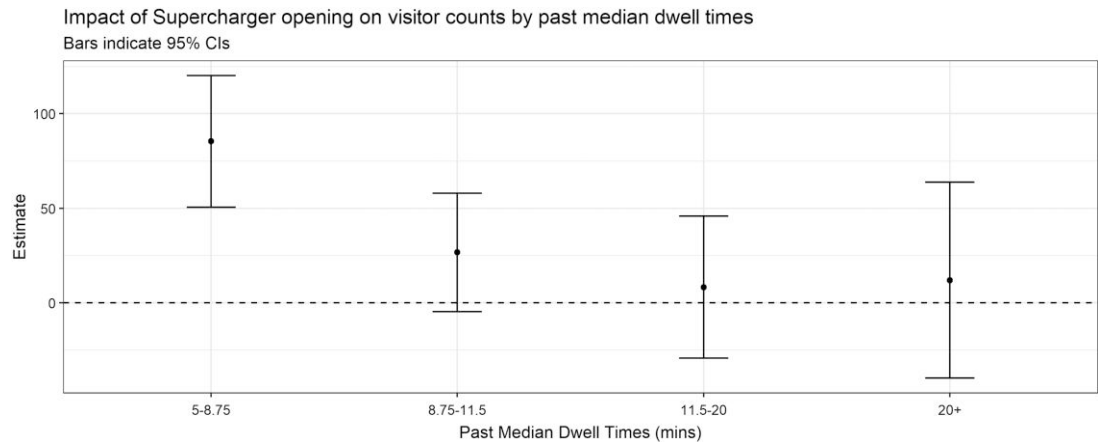
Next, we consider the importance of available parking spots. One fear a retail establishment might have with giving up parking space for a charger is the potential that it may displace existing, non-Tesla visitors. With this in mind, we might expect to see stronger positive effects on visit volumes when a Supercharger is placed in a larger parking lot. To test this idea, we use OSM’s car and vehicle parking geospatial data to calculate the *ParkingArea* of the lot (demeaned, in 1,000s of square

feet). Importantly, not all Superchargers are identified as being located in a publicly accessible parking lot, according to OSM. This might occur, for example, if a Supercharger is located in a paid parking garage or an open lot in a strip mall. Accordingly, we first include a dummy indicator of whether the Supercharger was identified as located in a parking lot according to OSM. We then incorporate a measure of square footage for said lots. Once again, these measures are undefined for control locations, and the latter measure is undefined for locations not positioned in an OSM-tagged parking lot. Fortunately, we can impute any value we wish, as the main effect of either variable and their interaction will be subsumed by our establishment fixed effects. As expected, for those Superchargers tagged as located in a parking lot by OSM, that is, where a parking lot size is available, we see larger positive impacts on establishment visits as the lot size increases ($\beta = 1.18, p < 0.05$).

Next, we flexibly examine the relationship between the physical distance between a Supercharger and the retail establishment, and we flexibly consider the importance of a typical service encounter duration. We introduce a vector of dummies as moderators for each of these estimations, capturing ranges of distance or duration for a typical service encounter (dwell time) at each establishment pretreatment.

We begin with the visit duration analysis, introducing quartile dummies based on historical median dwell times. We present the regression results in the online appendix in Table EC.11, and we graphically depict the coefficients in Figure 6. We find that the benefits for retailers are much larger for establishments where service encounters are relatively short. This result is consistent with the idea that establishment service offerings need to be aligned,

Figure 6. Impact of Supercharger on Visitor Counts for Top-Mentioned Brands, Matched Panel Within 200 Meters of a Rapid Charger by Quantiles of Past Median Dwell Times



temporally, with the duration of a typical charging session (e.g., less than 20–25 minutes) for the establishment to benefit meaningfully in terms of increased foot traffic.

Considering flexible distance effects, we replace the binary treatment dummy with a series of dummies based on ranges of distance (in increments of 150 meters) from the treating Supercharger, up to 750 meters.⁴ The results of this estimation are presented in Figure 5, and the tabular results are presented in Table EC.12 of the online appendix. We find that significant visit benefits manifest only at less than 150 meters from the charger. Notably, this result is consistent with the results reported earlier using our continuous moderator. This finding is also consistent with our initial treatment definition, based on a 200-meter radius around each Supercharger. Notably, we explore these effects further in Figure EC.5 and Table EC.13 of the online appendix, considering even more granular distance increments of just 30 meters, using our primary panel of top-mentioned brands. Those estimates further indicate that the majority of demand spillovers accrue within just 60 meters.

Finally, we consider heterogeneity in the effects across different types of retail establishments.⁵ We constructed establishment-type dummies based on groupings of NAICS codes, which we interact with the treatment indicator. The NAICS code groupings that we employ are detailed in the online appendix in Table EC.14. We report coefficients both numerically and graphically in Figure EC.6 and Table EC.15 of the online appendix. Furthermore, Figure EC.7 of the online appendix depicts effect estimates by subcategories of retail shops. Broadly, we find that the positive effect of a Supercharger on retailer visits primarily manifests among fast food restaurants, convenience stores, grocers, and bookstores, with a weakly positive effect on full-service (sit-down) restaurants. These results thus align with our other results, which indicate the importance of temporal alignment

between an establishment’s typical service encounter duration and the duration of a typical charging session.

5. Discussion and Implications

We have presented a large-scale empirical study of the effects of EV charging stations on the demand experienced by proximate brick-and-mortar retail establishments. These results support EV network operators’ claims that hosting an EV charging station can provide retail establishments with access to a new, desirable consumer audience. Retailers gain spillover demand by virtue of their synergistic colocation with EV charging stations. Our estimates indicate an approximate 4% rise in foot traffic, on average.

Assuming transactions are of homogenous value in terms of revenue, our estimate of the average effect on visit volumes translates to an approximate increase of \$0.17 per transaction for an average convenience/quick grocer (Courier Post: How Wawa Makes Money). In the case of a true grocery store, like Whole Foods, our effects equate to an additional \$1.84 (Source: Business Insider) per transaction. Notably, our estimates of consumer spending, based on the subsample of spending data we have access to (Table 3), are consistent with the assumption; we see that the rise in visits also translates to a roughly comparable rise in revenue.

The effect estimates we report depend on the number of Teslas operating in a given location (300,000 Teslas sold in 2021, which remain relatively limited compared with gas-powered vehicles about 15 million vehicles overall). Nationwide, on average, there are 14 EVs per charger outlet (based on state-level statistics), whereas the ratio of gas-powered vehicles to gas pumps is much, much higher, approximately 1,600:1 (assuming an average of 12 pumps per each of the 145,000 retail gas outlets and 276 million gasoline vehicles). As the number of EVs on the road continues to rise, retailers might expect to experience larger gains going forward.

Our findings, particularly those related to sources of heterogeneity, provide important insights for retailers about whether and when they may expect to benefit from consumer demand spillovers from EV charging infrastructure. For example, our results indicate the rapid decay in benefits with physical distance from a charger. As EV network operators are unlikely to install redundant charging options in a single geographic area, this finding suggests that it would be important for retailers to move early if they hope to benefit from EV charger hosting, as this would likely determine retailers' relative proximity to the physical charging station. Our estimates show that when the charger is located at a greater distance, even by just a few tens of meters, can make a major difference.

This consideration is compounded by the evidence we provide of first-mover advantages. We demonstrate that the first charging station infrastructure installed in a local market yields larger spillover demand benefits. This is plausibly explained by habit formation on the part of consumers, for example, consumers likely begin to use charging stations that are first made available and then continue to do so after they form a routine. This observation is important, as prior work notes the likely broader competitive impact among local retail establishments from investments in EV charging infrastructure (Arlt and Astier 2023). Our findings highlight that retailers should think carefully before engaging in any wait-and-see (free-riding) behavior, for example, in the hopes of free-riding on other proximate retailers' EV charger investments, because competitive concerns manifest at a larger geographic scale. Other charging options located elsewhere in the same geographic market (e.g., within the broader zip code or county), should they move first limit the potential upside for a later-adopting retail establishment.

Our findings also demonstrate that the benefits to a retailer may depend on the types of goods or services they provide. Our findings indicate that establishments offering services that typically require lengthier visits, for example, full-service restaurants (FSRs or sit-down restaurants) or department stores, receive lesser benefits from EV charging infrastructure. By contrast, convenience stores and grocery stores benefit systematically more. These results are likely tied to the fact that a restaurant visit tends to necessitate a relatively lengthier visit. In our sample, typical restaurant visits last approximately 19 minutes (14 minutes for fast food and 28 for full service), on average, compared with 14 minutes for grocery and 9 minutes for convenience stores (see Figure EC.8 in the online appendix). As such, retailers should be cognizant of the temporal synergies between the services they offer and a typical charging session, including the time walking to and from the store. More generally, our work speaks to the importance of context and synergies between a given retail establishment, in terms of the services it offers and the nature of the consumption

experience, and the complementarity of nearby service encounters that an establishment might hope to benefit from (in this case, an EV charging session). Beyond geographic proximity, we highlight the importance of understanding temporal synergies.

Our findings also have important implications for EV network operators. Most notably, our findings highlight the types of retailers and locations that EV owners are most likely to appreciate and avail themselves of as they wait for their vehicle to charge. To the extent an EV network operator can locate its stations in proximity to these ideal retailers, they are likely to see the greatest use and impact. Finally, our work bears implications for government policymakers, who are seeking to facilitate a more rapid expansion of EVs. Our findings suggest that government subsidy toward the expansion of EV networks may be better allocated toward areas of the country where retailer synergies are relatively lacking. Furthermore, governments might look to coordinate their efforts with major retail chains as they look to expand charging network infrastructure in the most cost-effective manner possible.

6. Limitations and Future Work

Our work is, of course, subject to several limitations. First, we consider only one charging network and EV manufacturer: Tesla. This fact may limit the generalizability of our results to some extent. Tesla Superchargers may differ systematically in some important respects. For example, other EV networks may differ in the number of outlets provided per station, the availability of app-based interfaces, and differing demographics among consumers of other EV types. However, we believe these should not be serious concerns for a few reasons. First, as noted earlier, the Tesla Supercharger network will soon be opened to all EV types; thus, differences in the features of Superchargers themselves may soon become irrelevant. Furthermore, Tesla owner demographics are broadly quite similar to the more general EV ownership demographic. That said, future work can explore the generalizability of our findings by considering additional EV charging networks, such as that operated by VW's subsidiary, Electrify America.

Second and last, our measure of foot traffic may fail to capture certain types of demand, drive-through transactions in particular. This is because our measures are location-based and dependent upon dwell times. Accordingly, our findings about spillover benefits to quick service restaurants (QSRs), that is, fast food locations, may be systematically understated. Future work might, therefore, look to employ alternative measures of demand, for example, sales metrics obtained directly from retail establishments.

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Endnotes

¹ We assessed the accuracy of these reports by randomly sampling 20 and manually identifying the first online mention of each in the news media, on other social media, or in Google Places reviews. We found that 70% of the sampled Superchargers were first mentioned in the same month as the installation reported in the Tesla owner forums. In general, these forum reports appeared at the same time as, or earlier than, the first mentions appearing in other online sources. Based on these observations, we concluded that the forum-reported opening dates are likely to be fairly accurate.

² Simple mention on the Tesla website does not necessarily imply a formal partnership existed between Tesla and a retailer. Many small, independent establishments were identified on Tesla's website. That said, some formal partnerships were indeed formed between Tesla and large retail chains, including Sheetz, Wawa, and Starbucks. Our main results are robust to excluding these brands from the estimation.

³ It should be noted that, when we replicated the matrix completion estimates using our expanded panel, described earlier in Section 4.3.2, we obtained the results presented in Figure EC.3 of the online appendix, which actually indicate that the effects persist into the long term (+20 months).

⁴ We necessarily leverage our expanded sample here (discussed in Section 4.3.2), which includes all retail brands and treated establishments located up to 750 meters from a Supercharger.

⁵ Once again, we use the expanded panel for this purpose to obtain greater representation across establishment types.

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